



Biventricular Loading During Ex-Vivo Heart Perfusion (EVHP) for Enhanced Functional and Metabolic Assessment of Donor Hearts



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Introduction: Limits in Assessing Heart Transplantability

Heart transplantation (HT) is the singular curative treatment for patients with end-stage heart disease. However, it remains limited due to the severe shortage of donor organs, underutilization of available organs, temporal and geographic constraints, and stringent listing criteria.

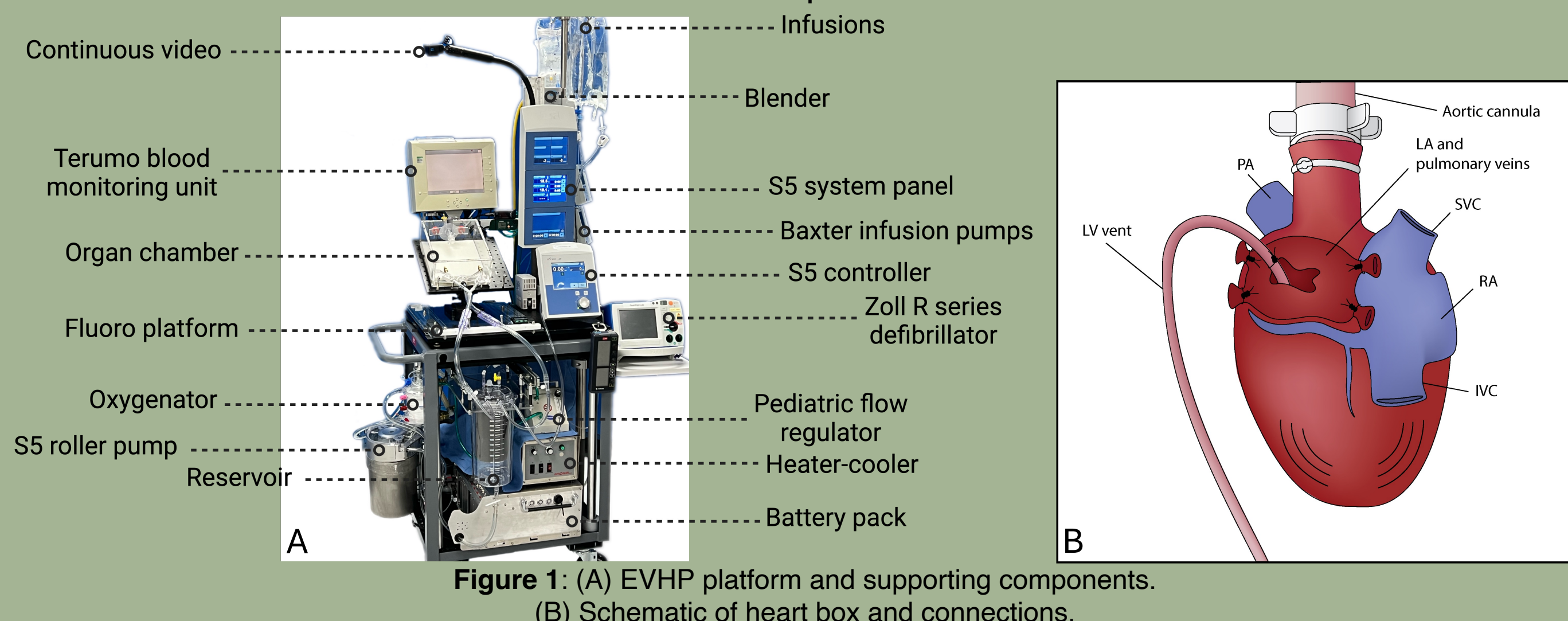
Notably, the use of hearts from donation after circulatory death (DCD) donors has increased the United States donor pool by 30%.¹ Before organ procurement, DCD donors must undergo withdrawal of life-sustaining support and circulatory arrest, which introduces a period of warm ischemia and precludes evaluation of function.

To address this, the strategy of ex-vivo perfusion (EVHP) has emerged to enable normothermic perfusion, reanimation, and assessment. Oskar Langendorff first performed isolated heart perfusion in the late 1800s.² The only clinically approved EVHP platform is the Transmedics Organ Care System (OCS™) (TransMedics, Inc., Andover, MA). Despite this progress, the ability to assess the heart on EVHP and determine transplantability is limited to lactate metabolism measurements and visual inspection.

The issue arises from the use of traditional Langendorff perfusion via retrograde aortic flow, where the ventricles remain empty. Consequently, this technique prevents assessment of valvular and contractile function, which is detrimental since animal models demonstrate a consistent 5-10% deterioration in chronotropic and contractile function per hour of perfusion.³ EVHP modification efforts include “passive afterload modes,” “pump supported modes,” and “left ventricular assist modes,”^{4,5} but none have been put into clinical practice. Cardiomyocytes require mechanical load for normal physiological function⁶ and cellular activity such as extracellular matrix synthesis.⁷ An unloaded heart loses ventricular interdependence via the Frank-Starling mechanism. As such, current EVHP models fail to adequately maintain normal heart function and allow for detailed evaluation.

Research Goal: Biventricular Working Heart System

The main aim was to adapt and refine our current EVHP system to enable employment of a loaded (“working heart”) perfusion mode in addition to its original unloaded (Langendorff) functionality while also implementing improved assessment and monitoring capabilities for real-time characterization of cardiac function and response.



Method: Swine Hearts for Optimization

To optimize and assess feasibility of circuit design iterations, swine hearts were used as a large animal model to serve as a suitable proxy for eventual evaluation of human hearts. This would further allow for simplification of the system set-up and development of a loading protocol.

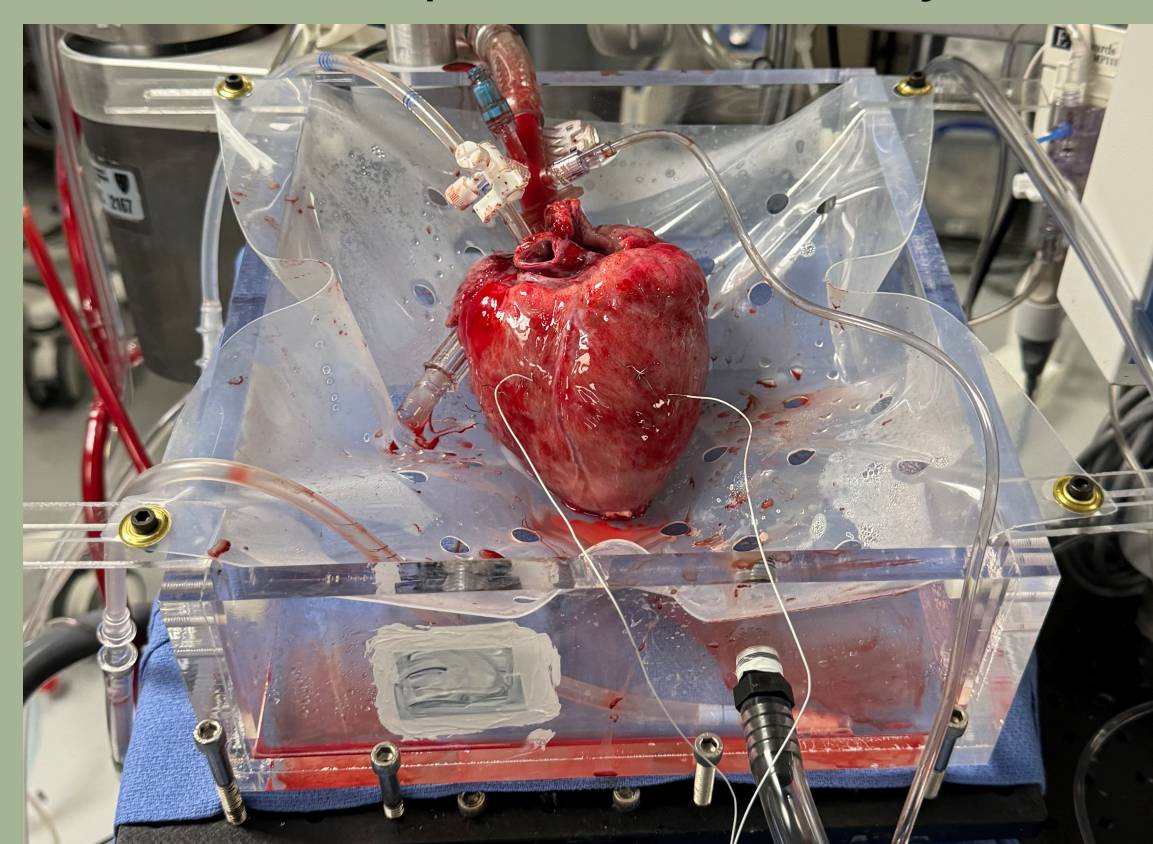


Figure 2: An unloaded swine heart in the EVHP heart chamber.

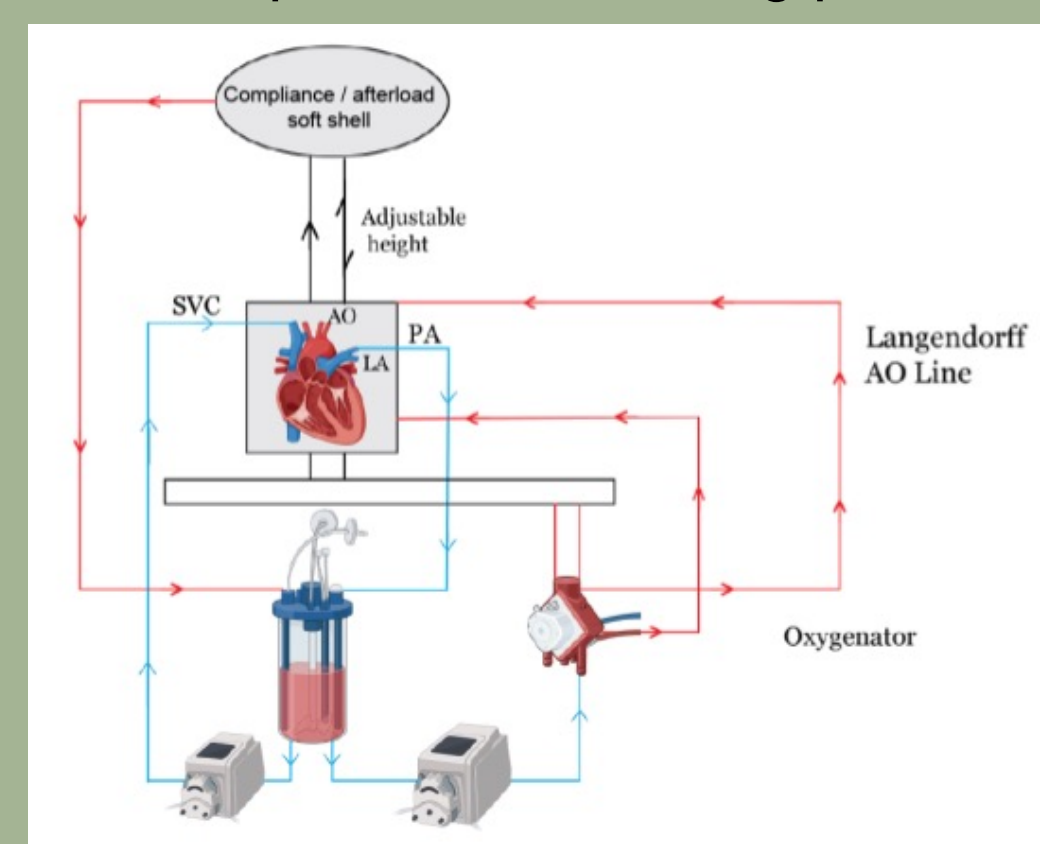


Figure 3: Biventricular loading circuit configuration.

Method: Computational Fluid Dynamics

In designing the initial circuit configuration, the need for custom and modified EVHP components were realized. Anticipated components included a rotatable aortic cannula and multi-function left atrial (LA) cannula to enable simultaneous inflow and left ventricular (LV) venting.

The designing of the cannulas was done using Fusion360 (Autodesk, Inc., San Francisco, CA). Designs were printed using an M2 3D printer (Carbon, Inc., Redwood City, CA) and Loctite IND405 Clear resin for prototyping of structural integrity and ease of use.

Computational fluid dynamics (CFD) analyses on the cannulas were performed to optimize flow and filling parameters along with reducing complexity and prime volume while enabling seamless transitions between loaded and unloaded states. Simulations were run and results were post-processed using Ansys Fluent (Ansys, Inc., Canonsburg, PA) with blood modeled as a non-Newtonian fluid based on the Carreau-Yasuda model. Post-processing focused on velocity pathlines, wall shear stress, and pressure.

$$\mu = \mu_{\infty} + \frac{(\mu_0 - \mu_{\infty})}{(1 + (\lambda\dot{\gamma})^a)^{\frac{1-n}{a}}}$$

Figure 4: Carreau-Yasuda equation used to model blood flow. μ is the effective viscosity, μ_{∞} (0.0035 kg/m·s) is the infinite shear viscosity, μ_0 (0.056 kg/m·s) is the zero shear viscosity, λ (3.13 s) is the time constant, n (0.3568) is the viscosity exponent, a (2) is the transition parameter, and $\dot{\gamma}$ is the effective shear rate. The density used was 1060 kg/m³.

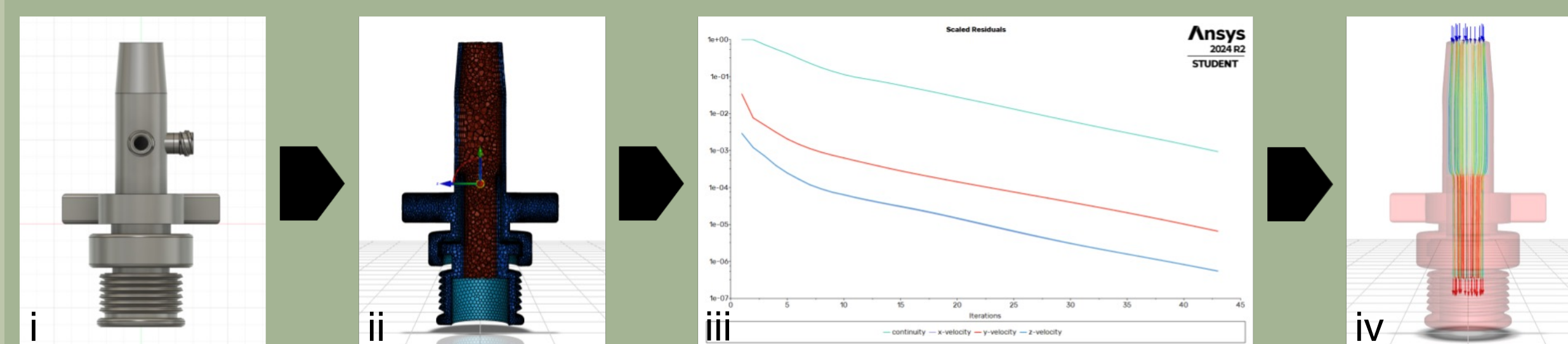


Figure 5: CFD analysis workflow. (i) Solid geometry set-up. (ii) Surface and volume meshing. (iii) Simulation set-up and run. (iv) Post-processing.

Results: Preserved Cardiomyocyte and Mitochondrial Structures on EVHP

In initial traditional EVHP validation trials (n = 6), biopsies demonstrated preserved cardiomyocyte and mitochondrial structures. Transmission electron microscopy (TEM) images revealed regularly spaced Z lines and mitochondria with no architectural distortion or membrane abnormalities. Hematoxylin and eosin (H&E) stains revealed regular cardiomyocyte membrane presence, minimal amounts of architectural distortion, and no signs of excessive interstitial edema.

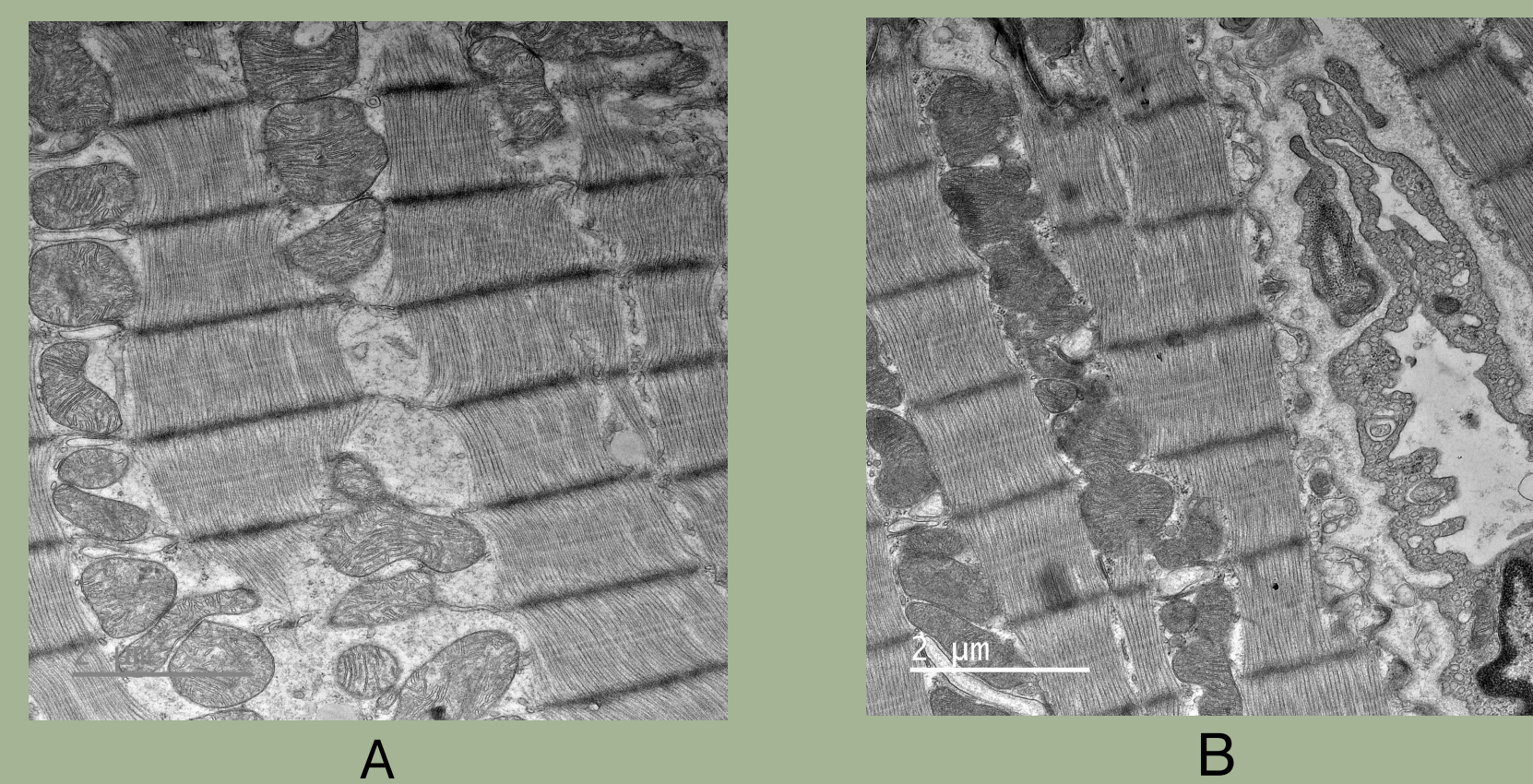


Figure 6: Representative TEM images at 6000x of LV samples. (A) After 4 hours of EVHP and a transplant. (B) Control.

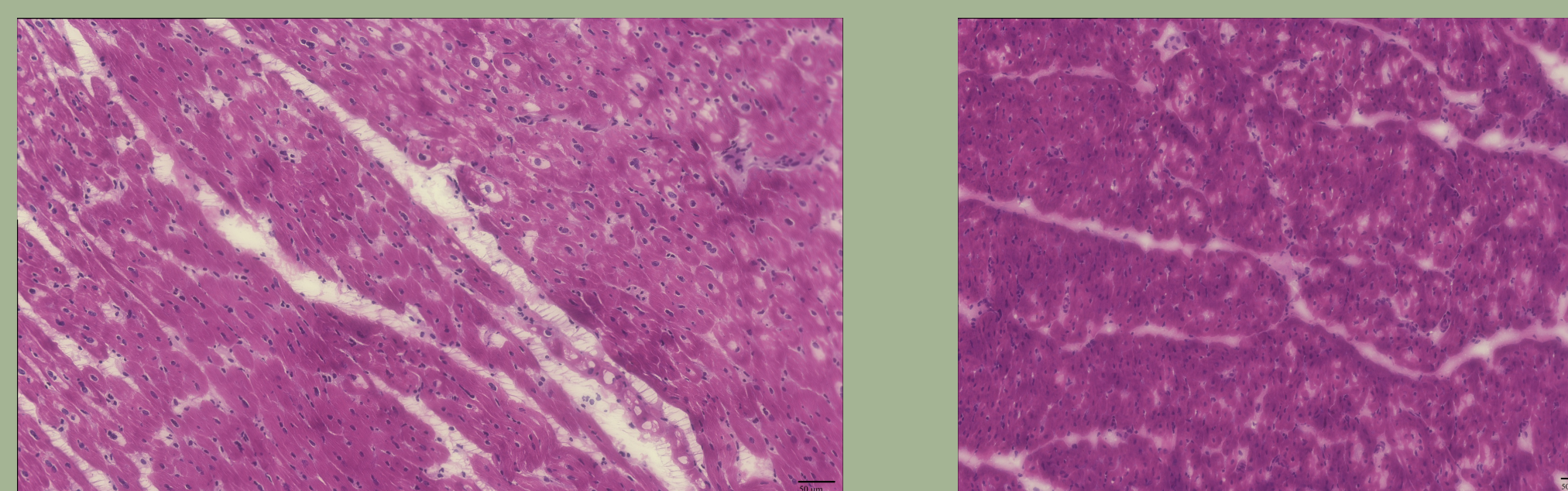


Figure 7: Representative H&E stains. (A) After 4 hours of EVHP and a transplant. (B) Control.

Results: Sustained Cardiac Stability on EVHP

In the EVHP validation trials, monitored values remained within acceptable limits with lactate values under 4 mmol/L for all trials.

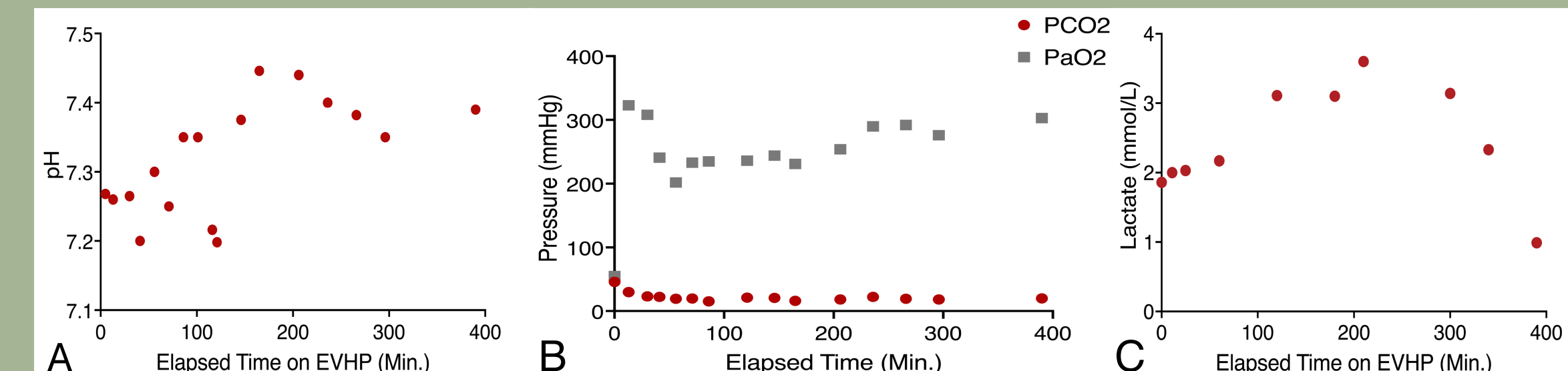


Figure 8: Representative trends of monitored values on EVHP (>6 hrs) demonstrating stability. (A) pH trend. (B) CO₂ and O₂ pressure trend. (C) Lactate trend.

Results: Increased Cardiac Contractility After Loading

Swine hearts (n = 2) were loaded in single or biventricular mode. In 5 to 10 minute intervals, loading conditions including chamber and flow parameters were adjusted for epicardial ultrasound and gross inspection.

After periods of loading, the hearts were observed unloaded and demonstrated increased contractility. Further functional analysis was not performed as the focus was on feasibility and initial protocol development. This preliminary data suggests a regimen of 30 minutes of rewarming and Langendorff perfusion followed by biventricular loading for 5 minute intervals every 30 minutes.

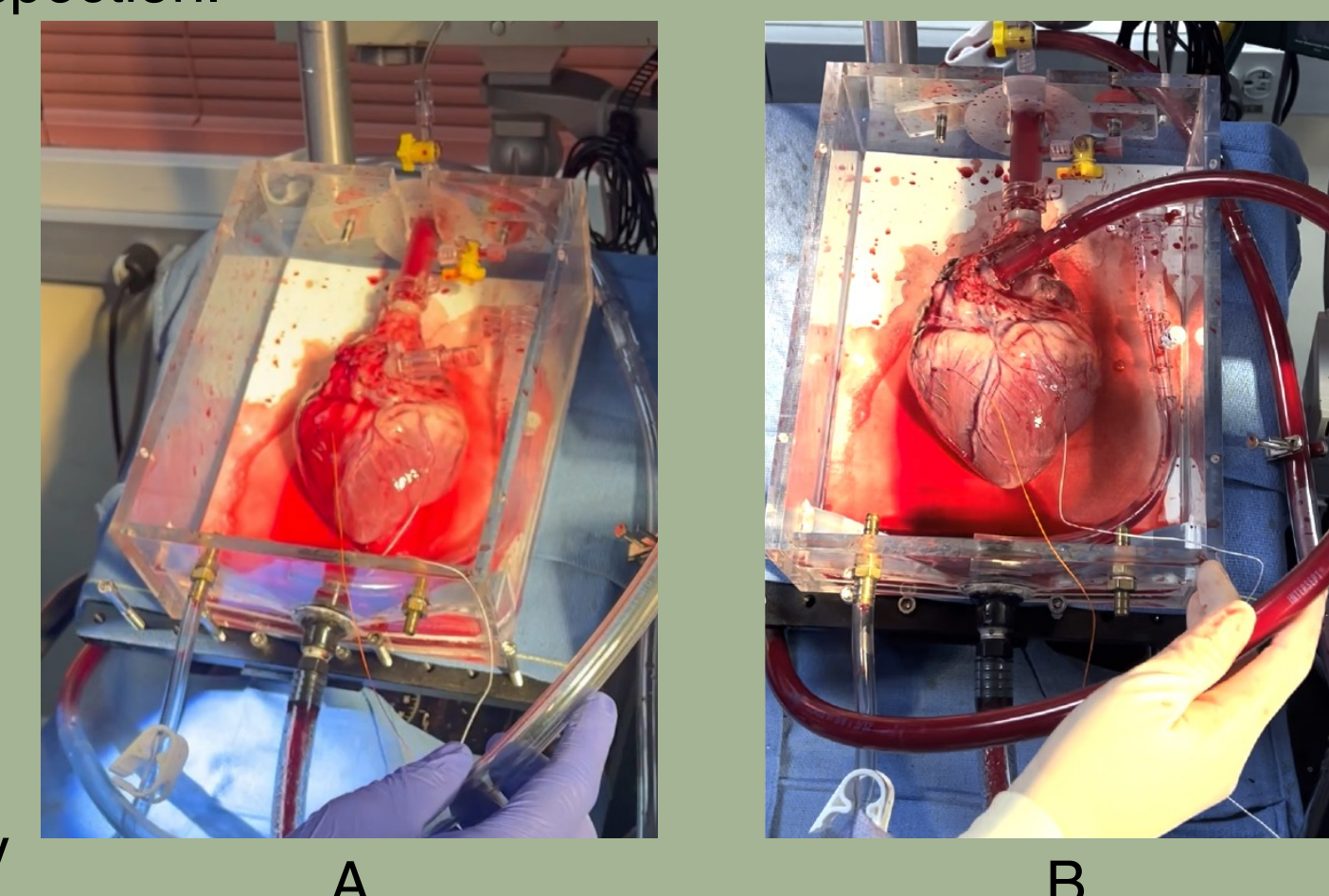


Figure 9: Single ventricular working heart trial. (A) Langendorff perfusion prior to loading. (B) Working heart perfusion for LV.

Results: Improved Fluid Dispersion and Blood Flow

We calculated the average Reynolds number (Re) for both cannulas using a viscosity of 0.00405 kg/m·s. The aortic cannula Re = 1423.4, and the LA cannula Re = 1801.02. Both numbers below 2000 indicate laminar flow within each, showing optimization for smooth blood flow and improved fluid dispersion. The cannulas have yet to be tested.

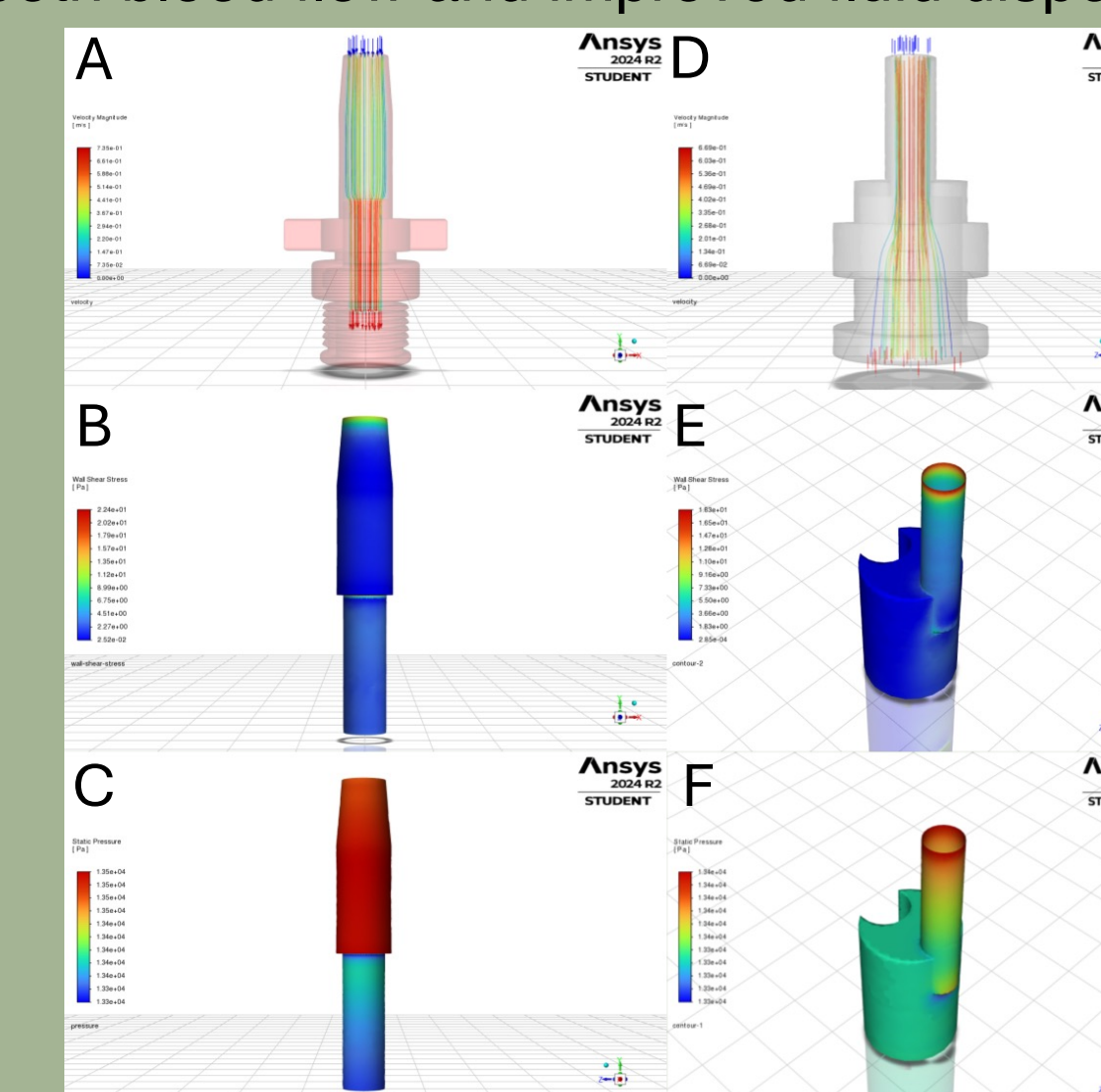


Figure 10: CFD simulation results for the aortic and LA cannula designs. (A) Aortic cannula velocity magnitude. (B) Aortic cannula wall shear stress. (C) Aortic cannula pressure. (D) LA cannula velocity magnitude. (E) LA cannula wall shear stress. (F) LA cannula pressure.

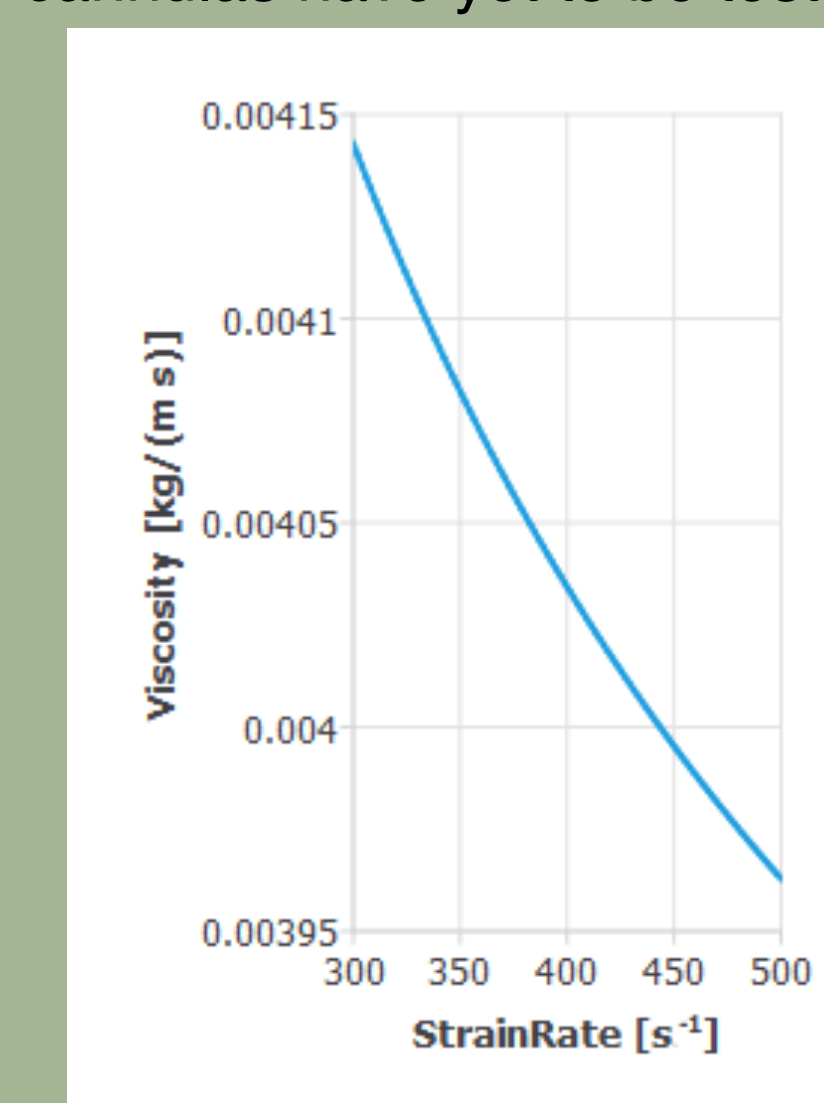


Figure 11: The calculated viscosities from strain rates of 300 s⁻¹ to 500 s⁻¹ by employing a shear rate dependent Carreau-Yasuda model using Ansys Fluent.

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